

Laplace Transform

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

06016401

MATHEMATICS FOR INFORMATION TECHNOLOGY

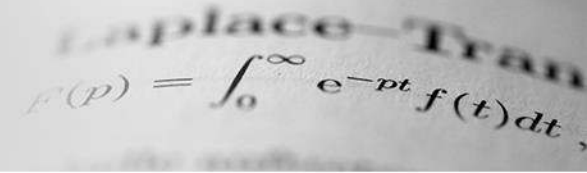
Instructor: Dr. Praphan Pavarangkoon

Office: 5th Floor

Email: praphan@it.kmitl.ac.th

Office hours: Thursday at 9:00 – 11:00 or as an advance appointment

Outline

A decorative header image showing a close-up of a document with the Laplace Transform formula. The text "Laplace-Trans" is visible at the top, and the formula $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$ is written below it.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- Course Introduction
 - Syllabus
 - Calendar
- Linear Algebra
- Matrices
 - Definition of a Matrix
 - Matrix Addition
 - Scalar Multiplication
 - Matrix Multiplication
 - The Transpose of a Matrix

Course Introduction

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- Course: 06016401
- Course Title: Mathematics for Information Technology
- Duration: 17 weeks (including Midterm and Final examination weeks)
- Credit Hours: 3 (3-0-6) credit hours
 - Lecture: 45 hours
 - Self-Study: 90 hours
- Lecturers
 - Asst. Prof. Dr. Somkiat Wangsiripitak
 - Dr. Praphan Pavarangkoon

Course Introduction (cont.)

- Teaching Assistants

- 60606002 ศิริพร ดอกทุเรียน (ตุ๊กตา)
- 61070238 สหสวรรณ ขันรักษา (ปอนด์)
 - Tuesday: 2 hours at 16:00 - 18:00
- 64070106 ศศิธร ศรีจันทร์ (ไหม)
- 64070197 พณิตา สายศร (แป้ง)
 - Thursday: 2 hours at 16:00 - 18:00
 - Friday: 2 hours at 16:00 - 18:00

- Course Date/Time/Room

- This class meets every Tuesday and Wednesday (3 hours/week).
 - Tuesday: 1.5 hours at 9:00 - 10:30, 10:30-12:00 and 12:00 - 13:30
 - Wednesday: 1.5 hours at 14:00 - 15:30, 15:30-17:00 and 17:00 - 18:30

Course Description



Laplace Transform
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- ลิมิตและความต่อเนื่อง อนุพันธ์ของฟังก์ชัน อนุพันธ์อันดับสูง การประยุกต์ของอนุพันธ์ ปัญหาค่าสูงสุดและต่ำสุดของฟังก์ชัน การอินทิเกรตและเทคนิคของการอินทิเกรต การประยุกต์ของการอินทิเกรต ฟังก์ชันสองตัวแปร อนุพันธ์ย่อย พีชคณิตเชิงเส้น เมทริกซ์และดีเทอร์มิแนนต์ ระบบสมการเชิงเส้น และการหาผลเฉลยของระบบสมการเชิงเส้น เวกเตอร์สเปซ การแปลงเชิงเส้น การแปลงเมทริกซ์ เมทริกซ์เชิงตั้งฉาก

Aims and Objectives

A decorative image in the top right corner showing a portion of a document with the Laplace Transform formula:
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$
 and the text "Laplace-Trans" above it.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- จุดมุ่งหมาย
 - เพื่อให้ผู้เรียนมีความรู้พื้นฐานทางคณิตศาสตร์ที่จำเป็นสำหรับการเรียนในวิชาต่างๆ เกี่ยวกับเทคโนโลยีสารสนเทศ
- วัตถุประสงค์
 - มีการประยุกต์ใช้ซอฟต์แวร์ที่เหมาะสมนอกเหนือจากการคำนวณแบบธรรมดา เพื่อให้ผู้เรียนสามารถประยุกต์ความรู้ที่ได้เรียนในการแก้ปัญหาทางเทคโนโลยีสารสนเทศได้

Topics and Details of 17 Weeks

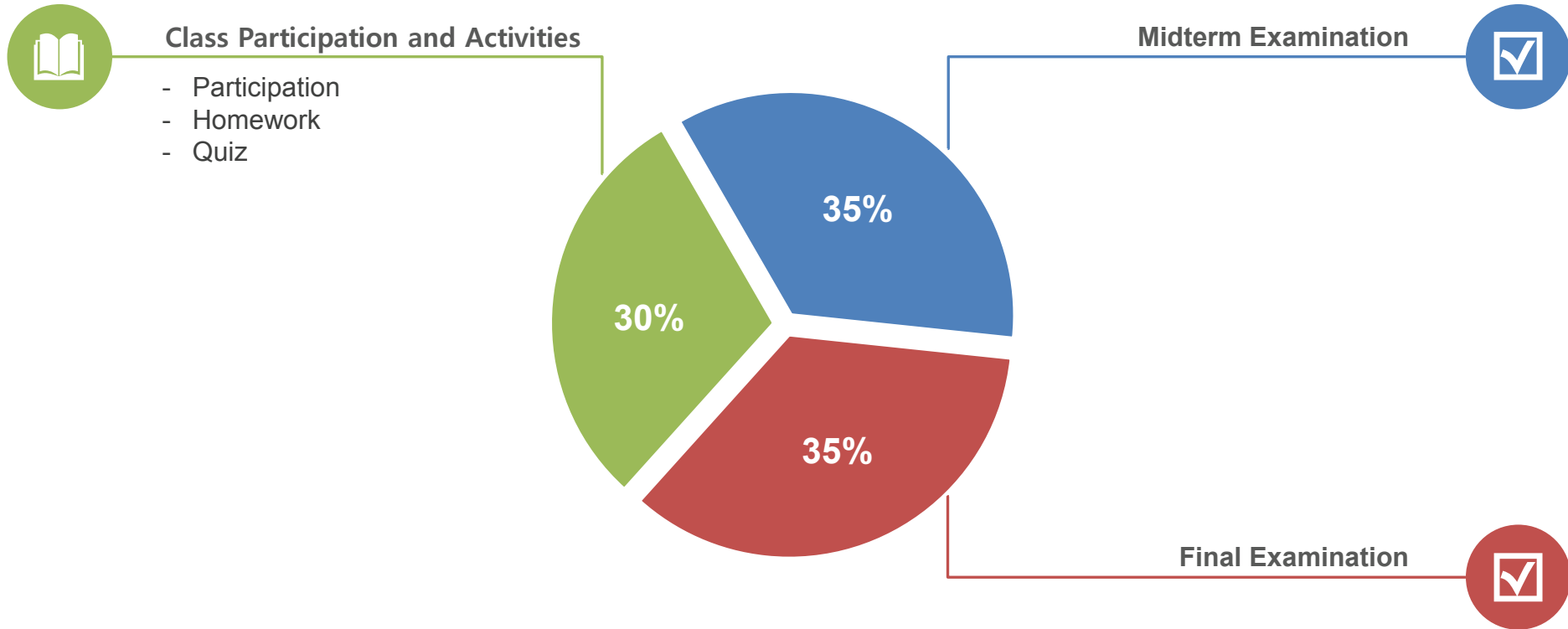
Week	Date	Topics and Details	Lecturer
1	02/08 and 03/08	Linear algebra	Praphan
2	09/08 and 10/08	Matrices and determinants	Praphan
3	16/08 and 17/08	System of linear equations and its solutions	Praphan
4	23/08 and 24/08	Vector space (1)	Praphan
5	06/09 and 07/09	Vector space (2)	Praphan
6	13/09 and 14/09	Linear transformations	Praphan
7	20/09 and 21/09	Orthogonal matrix	Praphan
8	Midterm examination Thursday 29 September at 13:30 – 16:30		

Topics and Details of 17 Weeks (cont.)

Week	Date	Topics and Details	Lecturer
9	04/10 and 05/10	Function, limit and continuity	Somkiat
10	11/10 and 12/10	Derivative of functions	Somkiat
11	18/10 and 19/10	Higher-order derivatives, chain rule, implicit differentiation	Somkiat
12	25/10 and 26/10	Applications of the derivative	Somkiat
13	01/11 and 02/11	Integration (indefinite & definite)	Somkiat
14	08/11 and 09/11	Integration by substitution	Somkiat
15	15/11 and 16/11	Applications of integration	Somkiat
16	22/11 and 23/11	Function of two variables, partial derivatives	Somkiat
17	Final examination Thursday 1 December at 13:30 – 16:30		

Course Evaluation and Evaluation Criteria

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$



Course Assessment

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

Description	Grade		Score (points)
Excellent	A	4.0	80-100
Very Good	B+	3.5	75-79
Good	B	3.0	70-74
Fairly Good	C+	2.5	65-69
Fair	C	2.0	60-64
Poor	D+	1.5	55-59
Very Poor	D	1.0	50-54
Fail	F	0.0	0-49

หมายเหตุ ผู้สอนอาจจะพิจารณาแบบอิงกลุ่มร่วมด้วย

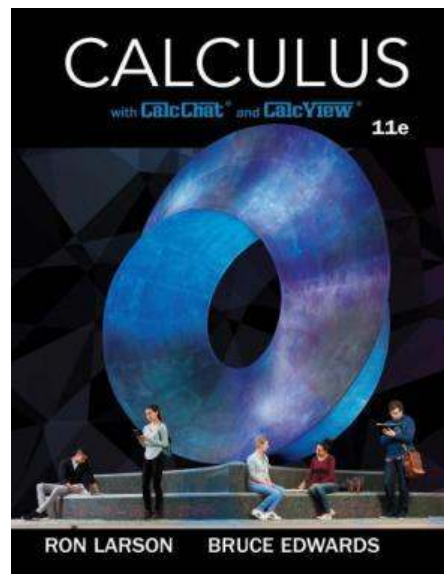
Resources

- ตำราและเอกสารหลัก

- Title: Calculus

Authors: Larson Edwards and Bruce H. Edwards

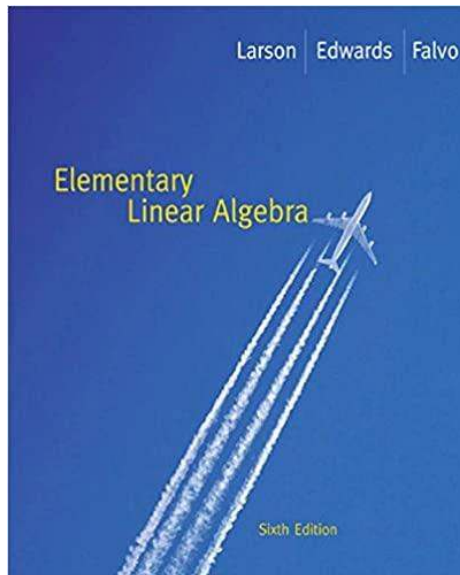
Publisher: Brooks/Cole, Cengage Learning



Resources (cont.)

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- Title: Elementary Linear Algebra
Authors: Ron Larson and David C. Falvo
Publisher: Houghton Mifflin Harcourt



Note

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- Student Responsibility, Course Policies (Attendance, Late/Absence/Medical Reasons), Disabilities and Accommodations, Academic Dishonesty (Plagiarism), and KMITL rules and regulations: Please refer to **Curriculum Guide**

What Is Linear Algebra?

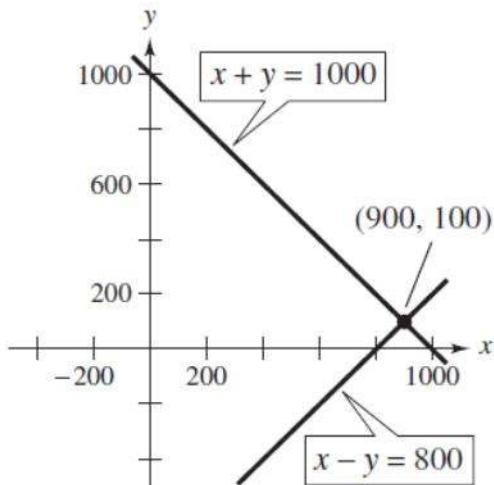
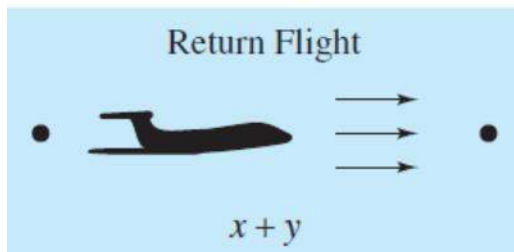
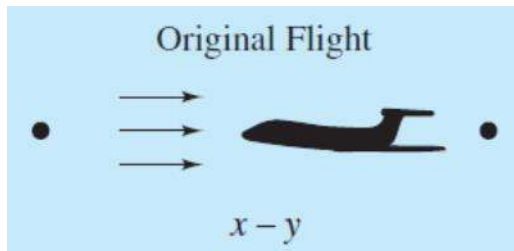
A decorative image in the top right corner showing a portion of a document with the Laplace Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$. The text "Laplace-Trans" is visible above the formula.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- The most fundamental theme of linear algebra is the theory of systems of linear equations.
- You have probably encountered small systems of linear equations in your previous mathematics courses.
- For example, suppose you travel on an airplane between two cities that are 5000 kilometers apart.
- If the trip one way against a headwind takes 6.25 hours and the return trip the same day in the direction of the wind takes only 5 hours, can you find the ground speed of the plane and the speed of the wind, assuming that both remain constant?

What Is Linear Algebra?

(cont.)

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$



The lines intersect at (900, 100).

- If you let x represent the speed of the plane and y the speed of the wind, then the following system models the problem.
$$6.25(x - y) = 5000$$
$$5(x + y) = 5000$$
- This system of two equations and two unknowns simplifies to
$$x - y = 800$$
$$x + y = 1000.$$
- and the solution is $x = 900$ kilometers per hour and $y = 100$ kilometers per hour.
- Geometrically, this system represents two lines in the xy -plane.
- You can see in the figure that these lines intersect at the point (900, 100) which verifies the answer that was obtained.

What Is Linear Algebra? (cont.)

A decorative image in the top right corner showing a portion of a document with the Laplace Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$. The text "Laplace-Trans" is visible above the formula.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- Solving systems of linear equations is one of the most important applications of linear algebra.
- It has been argued that the majority of all mathematical problems encountered in scientific and industrial applications involve solving a linear system at some point.
- Linear applications arise in such diverse areas as engineering, chemistry, economics, business, ecology, biology, and psychology.

Linear Algebra

A decorative image in the top right corner showing a blurred mathematical formula for the Laplace Transform: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- **Linear algebra** is a fairly extensive subject that covers vectors and matrices, determinants, systems of linear equations, vector spaces and linear transformations, eigenvalue problems, and other topics.
- As an area of study it has a broad appeal in that it has many applications in engineering, physics, geometry, computer science, economics, and other areas.
- It also contributes to a deeper understanding of mathematics itself.

Definition of a Matrix

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- A **matrix** is a rectangular array of numbers or functions which we will enclose in brackets.
- The plural of matrix is **matrices**.

If m and n are positive integers, then an $m \times n$ **matrix** is a rectangular array

$$\underbrace{\left[\begin{array}{ccccc} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{array} \right]}_{n \text{ columns}} \left. \vphantom{\begin{array}{c} a_{11} \\ a_{21} \\ a_{31} \\ \vdots \\ \vdots \\ a_{m1} \end{array}} \right\} m \text{ rows}$$

in which each **entry**, a_{ij} , of the matrix is a number. An $m \times n$ matrix (read “ m by n ”) has m **rows** (horizontal lines) and n **columns** (vertical lines).

Examples of Matrices



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Each matrix has the indicated size.

(a) Size: 1×1

$$[2]$$

(b) Size: 2×2

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

(c) Size: 1×4

$$\left[1 \quad -3 \quad 0 \quad \frac{1}{2} \right]$$

(d) Size: 3×2

$$\begin{bmatrix} e & \pi \\ 2 & \sqrt{2} \\ -7 & 4 \end{bmatrix}$$

Definition of a Matrix (cont.)

- One very common use of matrices is to represent systems of linear equations.
- The matrix derived from the coefficients and constant terms of a system of linear equations is called the **augmented matrix** of the system.
- The matrix containing only the coefficients of the system is called the **coefficient matrix** of the system. Here is an example.

System

$$\begin{array}{rcrcrcrcrl} x & - & 4y & + & 3z & = & 5 \\ -x & + & 3y & - & z & = & -3 \\ 2x & & & & -4z & = & 6 \end{array}$$

Augmented Matrix

$$\left[\begin{array}{cccc} 1 & -4 & 3 & 5 \\ -1 & 3 & -1 & -3 \\ 2 & 0 & -4 & 6 \end{array} \right]$$

Coefficient Matrix

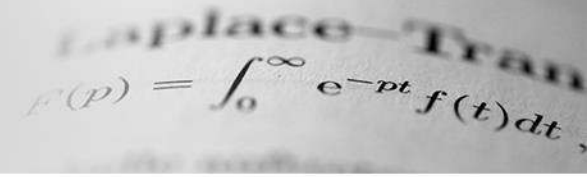
$$\left[\begin{array}{ccc} 1 & -4 & 3 \\ -1 & 3 & -1 \\ 2 & 0 & -4 \end{array} \right]$$

Operations with Matrices

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- Matrix Addition
- Scalar Multiplication
- Matrix Multiplication

Matrix Addition

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$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

- You can **add** two matrices (of the same size) by adding their corresponding entries.

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are matrices of size $m \times n$, then their **sum** is the $m \times n$ matrix given by

$$A + B = [a_{ij} + b_{ij}].$$

The sum of two matrices of different sizes is undefined.

Addition of Matrices



Laplace-Transform
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

(a) $\begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 1 & -2 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}$

(d) The sum of

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 0 & -1 \\ 3 & -2 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 1 \\ -1 & 3 \\ 2 & 4 \end{bmatrix}$$

Scalar Multiplication



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- When working with matrices, real numbers are referred to as **scalars**.
- You can multiply a matrix A by a scalar c by multiplying each entry in A by c .

If $A = [a_{ij}]$ is an $m \times n$ matrix and c is a scalar, then the **scalar multiple** of A by c is the $m \times n$ matrix given by

$$cA = [ca_{ij}].$$

You can use $-A$ to represent the scalar product $(-1)A$. If A and B are of the same size, $A - B$ represents the sum of A and $(-1)B$. That is,

$$A - B = A + (-1)B.$$

Subtraction of matrices

Scalar Multiplication and Matrix Subtraction

Laplace-Transform
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

For the matrices

$$A = \begin{bmatrix} 1 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} \quad \text{and} \quad -B = \begin{bmatrix} -2 & 0 & 0 \\ -1 & +4 & -3 \\ +1 & -3 & -2 \end{bmatrix}$$

find (a) $3A$, (b) $-B$, and (c) $3A - B$.

$$\begin{bmatrix} 3 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 6 & 12 \\ -10 & 4 & -6 \\ 7 & 0 & 4 \end{bmatrix}$$

Matrix Multiplication



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

If $A = [a_{ij}]$ is an $m \times n$ matrix and $B = [b_{ij}]$ is an $n \times p$ matrix, then the **product** AB is an $m \times p$ matrix

$$AB = [c_{ij}]$$

where

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{in}b_{nj}.$$

- This definition means that the entry in the i th row and the j th column of the product AB is obtained by multiplying the entries in the i th row of A by the corresponding entries in the j th column of B and then adding the results.

Finding the Product of Two Matrices

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Find the product AB , where

$$A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}.$$

3×2 3×2 2×2

$$\begin{bmatrix} 3 - 12 & -2 + 3 \\ -12 + 8 & 8 - 2 \\ -15 + 0 & 10 + 0 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ -4 & 6 \\ -15 & 10 \end{bmatrix}$$

Finding the Product of Two Matrices (cont.)

A close-up, slightly blurred image of a document showing the Laplace transform formula. The text "Laplace-Trans" is visible at the top, followed by the equation $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

Matrix Multiplication (cont.)

- Be sure you understand that for the product of two matrices to be defined, the number of columns of the first matrix must equal the number of rows of the second matrix. That is,

$$\begin{array}{ccc} A & B & = AB. \\ m \times n & n \times p & m \times p \\ \uparrow \quad \uparrow \quad \uparrow & & \\ \text{equal} & & \\ \text{size of } AB & & \end{array}$$

- So, the product BA is not defined for matrices such as A and B in the previous example.

Matrix Multiplication (cont.)

- The general pattern for matrix multiplication is as follows. To obtain the element in the i th row and the j th column of the product AB , use the i th row of A and the j th column of B .

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & a_{i3} & \cdots & a_{in} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2p} \\ b_{31} & b_{32} & \cdots & b_{3j} & \cdots & b_{3p} \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nj} & \cdots & b_{np} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1j} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2j} & \cdots & c_{2p} \\ \vdots & \vdots & & \vdots & & \vdots \\ c_{i1} & c_{i2} & \cdots & c_{ij} & \cdots & c_{ip} \\ \vdots & \vdots & & \vdots & & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mj} & \cdots & c_{mp} \end{bmatrix}$$

$a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{in}b_{nj} = c_{ij}$

Matrix Multiplication

Laplace-Transform
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

$$(a) \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -2 \end{bmatrix} \begin{bmatrix} -2 & 4 & 2 \\ 1 & 0 & 0 \\ -1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} -2+0-3 & 4+0+3 & 2+0-3 \\ -4-1+2 & 6+0-2 & 4+0-2 \end{bmatrix} = \begin{bmatrix} -5 & 7 & -1 \\ -3 & 4 & 2 \end{bmatrix}$$

2×3 3×3 2×3

$$(b) \begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3+0 & 4 \\ -2+0 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix}$$

2×2 2×2 2×2

$$(c) \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1+2 & 2-2 \\ -1+1 & 2-1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2×2 2×2 2×2

$$(d) \begin{bmatrix} 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+2-3 \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix}$$

1×3 3×1 1×1

$$(e) \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} \end{bmatrix}$$

3×1 1×3 3×3

Properties of Matrix Operations

- This section begins to develop the **algebra of matrices**.
- You will see that this algebra shares many (but not all) of the properties of the algebra of real numbers.

Properties of Matrix Addition and Scalar Multiplication

If A , B , and C are $m \times n$ matrices and c and d are scalars, then the following properties are true.

1. $A + B = B + A$
2. $A + (B + C) = (A + B) + C$
3. $(cd)A = c(dA)$
4. $1A = A$
5. $c(A + B) = cA + cB$
6. $(c + d)A = cA + dA$

Commutative property of addition

Associative property of addition

Associative property of multiplication

Multiplicative identity

Distributive property

Distributive property

Properties of Matrix Multiplication

If A , B , and C are matrices (with sizes such that the given matrix products are defined) and c is a scalar, then the following properties are true.

1. $A(BC) = (AB)C$
2. $A(B + C) = AB + AC$
3. $(A + B)C = AC + BC$
4. $c(AB) = (cA)B = A(cB)$

Properties of the Identity Matrix



Laplace Transform
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

- You will now look at a special type of square matrix that has 1's on the main diagonal and 0's elsewhere.

$$I_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

For instance, if $n = 1, 2$, or 3 , we have

$$I_1 = [1], \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

1×1 2×2 3×3

- When the order of the matrix is understood to be n , you may denote I_n simply as I .

Properties of the Identity Matrix (cont.)

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$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

If A is a matrix of size $m \times n$, then the following properties are true.

1. $AI_n = A$
2. $I_m A = A$

As a special case of this theorem, note that if A is a *square* matrix of order n , then

$$AI_n = I_n A = A.$$

The Transpose of a Matrix

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

The **transpose** of a matrix is formed by writing its columns as rows. For instance, if A is the $m \times n$ matrix shown by

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix},$$

Size: $m \times n$

then the transpose, denoted by A^T , is the $n \times m$ matrix below

$$A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} & \cdots & a_{m1} \\ a_{12} & a_{22} & a_{32} & \cdots & a_{m2} \\ a_{13} & a_{23} & a_{33} & \cdots & a_{m3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & a_{mn} \end{bmatrix}.$$

Size: $n \times m$

The Transpose of a Matrix (cont.)

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Find the transpose of each matrix.

(a) $A = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$ (b) $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ (c) $C = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (d) $D = \begin{bmatrix} 0 & 1 \\ 2 & 4 \\ 1 & -1 \end{bmatrix}$

$$\begin{bmatrix} 2 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 & 1 \\ 1 & 4 & -1 \end{bmatrix}$$

Properties of Transposes



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

If A and B are matrices (with sizes such that the given matrix operations are defined) and c is a scalar, then the following properties are true.

1. $(A^T)^T = A$ Transpose of a transpose
2. $(A + B)^T = A^T + B^T$ Transpose of a sum
3. $(cA)^T = c(A^T)$ Transpose of a scalar multiple
4. $(AB)^T = B^T A^T$ Transpose of a product

The Product of a Matrix and Its Transpose

For the matrix

$$A = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -2 & -1 \end{bmatrix}_{3 \times 2}$$

$$A^T = \begin{bmatrix} 1 & 0 & -2 \\ 3 & -2 & -1 \end{bmatrix}_{2 \times 3}$$

find the product AA^T and show that it is symmetric.

3×3

$$\begin{bmatrix} 1+9 & 0-6 & -2-3 \\ 0-6 & 0+4 & 0+2 \\ -2-3 & 0+2 & 4+1 \end{bmatrix} = \begin{bmatrix} 10 & -6 & -5 \\ -6 & 4 & 2 \\ -5 & 2 & 5 \end{bmatrix}$$

Q & A

A close-up photograph of a document showing the Laplace Transform formula. The text "Laplace-Trans" is visible at the top, followed by the equation $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$ and the phrase "with the condition" below it.

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$
with the condition

Q & A

A close-up photograph of a document showing the Laplace Transform formula. The text "Laplace-Trans" is visible at the top, followed by the equation $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$ and some faint, illegible text below it.

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$
with initial...